

Dataset before and after preprocessing

1- Data Cleaning:

1.1 - Duplicates: 34 duplicate rows are removed.

Before:

```
➤ print(df.shape)

df.info()

print(df.isnull().sum())

print(df.duplicated().sum())

print(df.dtypes)

(740, 21)
```

After:

```
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)

Shape after removing duplicates: (706, 21)
```

1.2 - Outliers: IQR used to remove outliers from these columns: distance_from_residence_to_work, age, work_load_average/day and absenteeism_time_in_hours.

Before:

```
➤ print(df.shape)

df.info()

print(df.isnull().sum())

print(df.duplicated().sum())

print(df.dtypes)

(740, 21)
```

After:

```

numeric_cols = ['distance_from_residence_to_work', 'age', 'work_load_average/day', 'absenteeism_time_in_hours']

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]

print("Shape after removing outliers:", df.shape)

```

Shape after removing outliers: (628, 21)

1.3 - Column Formatting: convert all column names to lowercase and replace spaces with underscores.

Before:

	ID	Reason for absence	Month of absence	Day of the week	Seasons	Transportation expense	Distance from Residence to Work	Service time	Age	Work load Average/day	...	Disciplinary failure	Education	Son	Social drinker	Social smoker	Pet	We
0	11	26	7	3	1	289	36	13	33	239.554	...	0	1	2	1	0	1	
1	36	0	7	3	1	118	13	18	50	239.554	...	1	1	1	1	0	0	
2	3	23	7	4	1	179	51	18	38	239.554	...	0	1	0	1	0	0	
3	7	7	7	5	1	279	5	14	39	239.554	...	0	1	2	1	1	0	
4	11	23	7	5	1	289	36	13	33	239.554	...	0	1	2	1	0	1	

5 rows × 21 columns

After:

	id	reason_for_absence	month_of_absence	day_of_the_week	seasons	transportation_expense	distance_from_residence_to_work	service_time	age	work
0	11		26	7	3	1	289	36	13	33
1	36		0	7	3	1	118	13	18	50

1.4 - Missing Values: No missing values.